"Satyam" → String

Operators & Conditionals (Logic)

± OR, AND

What are operators?

→ operators are symbols or, keywords that perform an operation on values.

5 + 3 → operands

4+3

Operands ← 5, 3

+ → operators

→ Unary operators → 1 operand

→ Binary " → 2 "

$\boxed{\text{typeof (variable\_name or, value)}}$

① Arithmetic operators → used for mathematical calculation.

+ → Addition

- → Subtraction

* → multiplication

/ → division

3 / 5 / 1

3 / 2 → %

/ → division
% → modulus (Remainder)   Remainder

** → Exponentiations → 2 ** 3 → 2 × 2 × 2 → 8
→ 5 ** 3 → 5 × 5 × 5 → 25 × 5 → 125

++ → increment
→ ++x → x = x + 1

-- → Decrement → --x → x = x - 1

Assignment operators →

x = 5 → Assignment operators.

let x = 5

+= → x += 2 → x = x + 2 → x = 7

-= → x -= 2 → x = x - 2 → x = 3

*= → x *= 2 → x = x * 2 → 10

/=

%=

Comparison operators

→ Return → true or, false

Type casting works here

→ == (equal) → value only

=== (Equal) → value & type

a = 5 $\quad$ $\boxed{a == b}$ → Type cast

b = '5' $\quad$ $\boxed{5 == 5}$ → True

!= not equal → =

!== Strict not equal

> Greather

< less than

>= Greator than equal to

<= less ______________

Logical operators

&& → Logical AND → All true → true

|| → Logical OR → Any true → true

! → Logical not → invorts ~~its bo~~

True → false

False → True

Bitwise operator

→ Bitwise operators treat theire operands as a sequence of bits (0, 1) rather than decimal. ↳ Binary

4 → decimal 32 16 8 4 2 1

$\boxed{100}$ → Binary

2^4 2^3 2^2 2^1 2^0

8 4 2 1

0 1 0 0

→ & — AND → Return 1 if both 1

5 & 3 →

0101
0011
―――――
0001 → 1

→ | → OR → either is 1 output 1

5 | 3

0101
0011
―――――

0111 → 7 ✓

~ → Bitwise not

↳ 5 → (~5) -(5+1)

$\boxed{-(5+1)}$ $\boxed{-(6)}$

1111 -

0101
1010 → -6

Left shift <<

→ shifts bits to the left

→ $\boxed{\text{multiplies the no. by 2}}$

5x2 = 10

a = 5

✓ 0101

$\boxed{01010}$ → 10

Right shift >>

↓ Right shif

→ divide by ~~half~~

→ every time you shift right by,

you can the num in half and drop remainder

2.5

↓ output

2

5 >> 1

0101 → 1 loss

010

0101 → (1) loss

0010 → (2)

Conditionals / Decision making

→ Conditionals allow JS to make decision

(i) if (ii) if else (iii) else if (iv) switch

(i) if

syntax -

```
if (condition) {     ← True
    // code
}
```

(ii) if else →

```
if (condition) {     (T) (F)
    ___
} else {
    // ___
}
```

(iii) else if

```
- if (condition) {
      // code
  } else if (condition) {
      // code
  }
  else if (condition) {
      // code
  }
  else {
      // code
  }
```

if else
↓
if else
↓
if else

→ **Switch** → switch is another way to handle multiple specific possible values.

Syntax -

```
switch (__) {
    case 1 "__" :
        ==
        ==
    __ - "__" :
```

case 2 "___" :

}default :

}

Ternary operator -

→ The ternary opertor is a short way of writting a simple if...else conditions

Syntax -

{condition ? valueif true : valueiffalse}